

MOTION DETECTION ALARM

INTRODUCTION TO EMBEDDED SYSTEMS

System is a set of interrelated parts/components which are designed/developed to perform common tasks or to do some specific work for which it has been created. Embedded means including something with anything for a reason. Or simply we can say something which is integrated or attached to another thing. Now after getting what actual systems and embedded mean we can easily understand what are Embedded Systems.

Embedded system is a computational system that is developed based on an integration of both hardware and software in order to perform a given task. It can be said as a dedicated computer system has been developed for some particular reason. But it is not our traditional computer system or general-purpose computers, these are the Embedded systems that may work independently or attached to a larger system to work on a few specific functions. These embedded systems can work without human intervention or with little human intervention.

COMPONENTS OF EMBEDDED SYSTEMS

1. Hardware
2. Software
3. Firmware

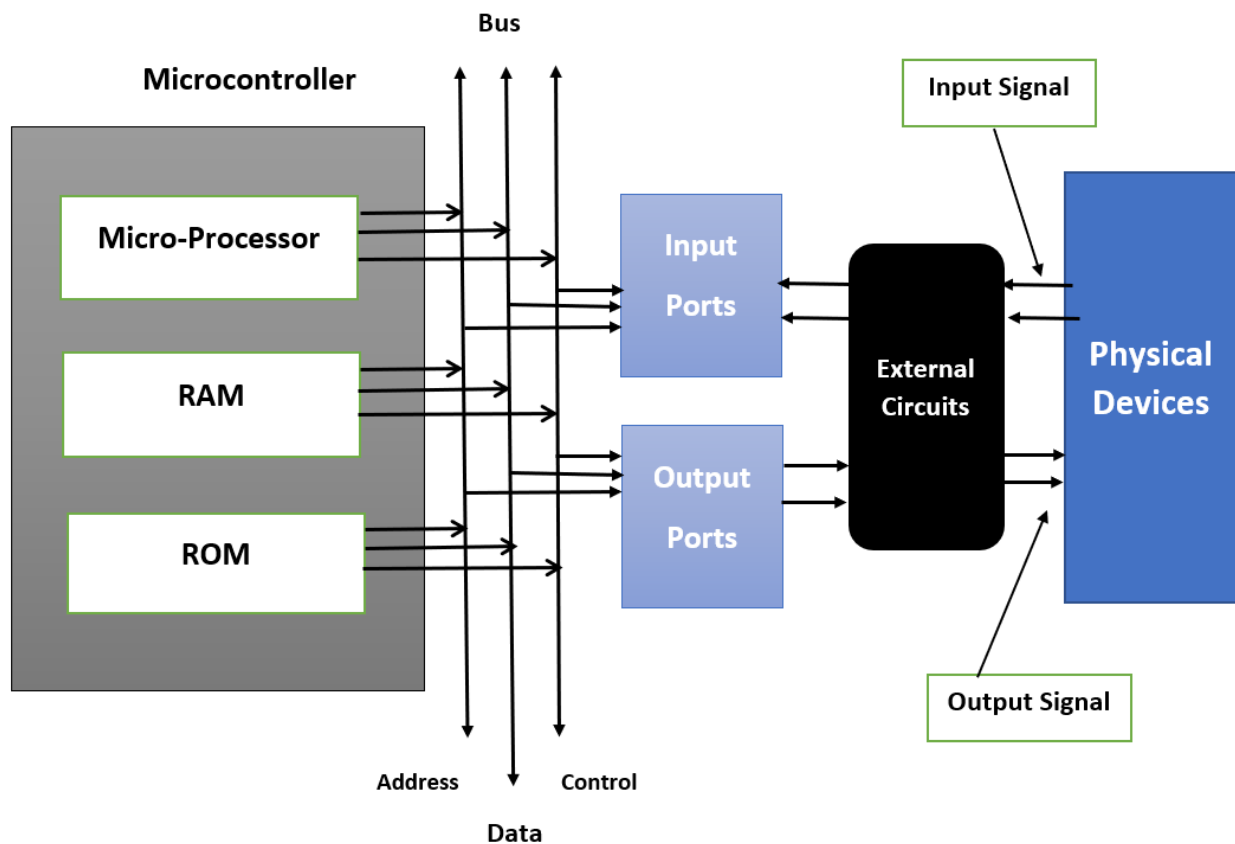
EXAMPLES OF EMBEDDED SYSTEMS

- Digital watches
- Washing Machine
- Toys
- Televisions
- Digital phones
- Laser Printer
- Cameras
- Industrial machines
- Electronic Calculators
- Automobiles
- Medical Equipment

APPLICATION OF EMBEDDED SYSTEM

- Home appliances
- Transportation
- Health care
- Business sector & offices
- Defense sector
- Aerospace
- Agricultural Sector

BLOCK STRUCTURE OF EMBEDDED SYSTEMS



ADVANTAGES OF EMBEDDED SYSTEM

- Small size.
- Enhanced real-time performance.
- Easily customizable for a specific application.

DISADVANTAGES OF EMBEDDED SYSTEM

- High development cost.
- Time-consuming design process.
- As it is application-specific less market available

HOW DOES AN EMBEDDED SYSTEM WORK?

Embedded systems operate from the combination of hardware and software that focuses on certain operations. An embedded system at its heart has microcontroller or microprocessor hardware on which user writes the code in form of software for control of the system. Here is how it generally works:

- **Hardware Layer:** Some of the hardware elements that are incorporated in an embedded system include the sensor, actuator, memory, current I/O interfaces as well as power supply. These components are interfaced with the micro controller or micro processor depending up on the input signals accepted.
- **Input/Output (I/O) Interfaces:** They to give the system input in form of data from sensors or inputs made by the users and the microcontroller processes the data received. The processed data is then utilized to coordinate the output devices such as displays, motors or communication modules.
- **Firmware:** Firmware which is integrated within a system's hardware comprises of certain instructions to accomplish a task. Such software is often used for real time processing and is tuned to work in the most optimal manner on the system hardware.
- **Processing:** Depending on the given software and the input data received from the system's inputs the microcontroller calculates the appropriate output or response and manages the system's components.
- **Real-time Operation:** Some of the most common systems are real time, this implies that they have the ability to process events or inputs at given time. This real time capability makes sure that the system accomplishes its intended function within stated time demands.

For instance therein an embedded system in a washing machine, the microcontroller would interface with the buttons (selections made by a user), sensors, for instance water levels, temperature and timers; it would control outputs such as motors, heaters and displays among others based on the program intended for washing cycles.

MINI PROJECT:

MOTION DETECTION SYSTEM USING ARDUINO:

The Motion Detection System is designed to detect the presence of motion using a PIR (Passive Infrared) sensor. When motion is detected, the system turns on an LED as an indication or alert.

COMPONENTS USED

- Arduino Uno
- PIR Motion Sensor
- Red LED
- 220 Ω Resistor
- Connecting wires

WORKING PRINCIPLE

- The PIR sensor continuously monitors infrared radiation in its surroundings.
- When a person or warm object moves within the sensor's range, the PIR sensor detects the change and sends a HIGH signal to **Digital Pin 2** of the Arduino.
- The Arduino processes this input and turns ON the LED connected to **Digital Pin 13** through a 220 Ω resistor.
- When no motion is detected, the sensor output becomes LOW and the LED turns OFF.

CIRCUIT CONNECTIONS

- PIR VCC $\rightarrow$ Arduino 5V
- PIR GND $\rightarrow$ Arduino GND
- PIR OUT $\rightarrow$ Arduino Digital Pin 2
- LED Anode $\rightarrow$ 220 Ω Resistor $\rightarrow$ Arduino Digital Pin 13
- LED Cathode $\rightarrow$ Arduino GND

APPLICATIONS

- Home security systems
- Automatic lighting
- Intruder detection
- Smart homes
- Office and warehouse monitoring

ADVANTAGES

- Low power consumption
- Simple and inexpensive design
- Easy

to

implement

SOURCE CODE:

```
// Pin definitions
const int PIR_PIN = 2;      // Signal pin from PIR sensor
const int ALARM_PIN = 13;   // Pin connected to LED and Buzzer

// Variables to track state
int pirState = LOW;         // Start assuming no motion detected
int val = 0;                // Variable for reading the pin status

void setup() {
  pinMode(PIR_PIN, INPUT);   // Declare sensor as input
  pinMode(ALARM_PIN, OUTPUT); // Declare alarm as output

  Serial.begin(9600);        // Initialize serial communication
  Serial.println("System Initialised. Monitoring for motion...");
}

void loop() {
  val = digitalRead(PIR_PIN); // Read input value

  if (val == HIGH) {          // Check if the input is HIGH
    digitalWrite(ALARM_PIN, HIGH); // Turn LED/Buzzer ON

    if (pirState == LOW) {
      // We have just turned on
      Serial.println("ALERT: Motion detected!");
      pirState = HIGH;        // Update the tracker variable
    }
  } else {
    digitalWrite(ALARM_PIN, LOW); // Turn LED/Buzzer OFF

    if (pirState == HIGH) {
      // We have just turned off
      Serial.println("Motion ended. System secure.");
      pirState = LOW;         // Update the tracker variable
    }
  }
  delay(100); // Small delay to improve simulation stability
}
```


SUMMARY:

This project is a Motion Detection System built using an Arduino Uno and a PIR sensor. The PIR sensor detects changes in infrared radiation caused by moving people. When motion is detected, it sends a signal to the Arduino through digital pin 2. The Arduino then turns on an LED connected to digital pin 13 through a 220-ohm resistor. When no motion is detected, the LED remains off. This system is commonly used in security alarms, automatic lighting, and smart home applications because it is simple, cost-effective, and energy efficient.